

# Anywhere Hyakuyobako: Construction and Application of Distributed Weather Observation System by BLE-Advertise

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## 1. Abstract

This study proposes "Anywhere-Hyohyobako," a distributed weather observation system that does not require any communication infrastructure. The system consists of autonomous sensor nodes that periodically transmit environmental data—such as temperature, humidity, and pressure—via Bluetooth Low Energy (BLE) advertising, and mobile devices or dedicated gateways that receive the data. Each node is powered by a solar panel and a rechargeable battery, enabling flexible deployment in diverse locations including urban areas, agricultural fields, and both indoor and outdoor environments without the need for power or internet access.

Furthermore, the system incorporates a method where images taken during sensor installation are analyzed using machine learning to estimate metadata that describes the installation environment. This approach helps ensure the reliability and accuracy of observed weather data, even under non-standard conditions. This paper outlines the system architecture, communication protocols, data acquisition and correction methods, and discusses its potential applications as a citizen-participatory observation network.

## 2. Introduction

Monitoring microclimates in urban environments is increasingly important for addressing issues such as heat island effects, heatstroke prevention, and disaster response. However, conventional weather observation systems often suffer from limitations in spatial resolution due to high installation costs and infrastructure requirements. While sensor networks that incorporate citizen participation are gaining popularity, challenges related to communication environments and power supply remain.

This study aims to overcome these limitations by developing an observation infrastructure that is easy to install and facilitates the aggregation and utilization of environmental data.

### 3. System Architecture and Features

The proposed system consists of the following three components:

- Sensor Node (M5Stamp Pico + ENV Unit): Equipped with periodic BLE advertising, autonomous power supply, and device ID functionality.
- Receiver Device (e.g., smartphones, ESP32, Raspberry Pi): Scans for BLE packets and forwards received data to the cloud via the internet.
- Cloud Platform (e.g., Supabase): Provides time-series data storage, spatial filtering, and a visualization dashboard.

Since the system operates solely on BLE proximity communication without requiring any additional network infrastructure, it can function even in remote areas or locations without cellular coverage. In addition, users can participate in the observation network simply by installing a mobile app, making the system highly accessible compared to conventional solutions.

### 4. Environmental Mapping via Image Analysis

The installation environment significantly affects sensor readings. To address this, the system includes a machine learning-based analysis feature that uses images taken during installation to estimate contextual metadata. The following environmental attributes are automatically inferred:

- Indoor / Outdoor
- Ground / Wall / Elevated
- Presence or absence of direct sunlight
- Surface material (estimated)

The inferred metadata is stored alongside the observation data and is used for correction during analysis. This enables consistent and statistically sound interpretations even under non-standard installation conditions.

### 5. Data Structure and Spatial Representation

The observation data are recorded in a structured format that includes sensor ID, timestamp, location, and environmental metadata. The location information is encoded using the H3 spatial index system, allowing aggregation at uniform spatial resolutions. This supports urban-

scale heatmap visualizations and time-based trend analysis.

Moreover, external access is supported via an API, enabling the data to be used in statistical analysis or as inputs for machine learning models.

## 6. Integration with Citizen Science and Applications

The system requires no specialized equipment or knowledge, allowing easy participation from citizens. In educational settings, it can serve as a hands-on learning tool that covers everything from sensor assembly and deployment to data visualization and analysis. It can also raise public awareness about local environmental issues.

From a research perspective, the system can be applied to studies of urban microclimates, high-resolution spatial interpolation of weather patterns, and real-time disaster monitoring.

## 7. Future Work

Although the current system focuses on temperature, humidity, and atmospheric pressure, future enhancements are planned as follows:

- Integration of additional sensors (e.g., light intensity, CO<sub>2</sub> concentration)
- Improved environmental mapping through advanced machine learning models
- International deployment targeting disaster-prone or infrastructure-limited regions

## 8. Conclusion

This paper presented the concept and implementation of "Anywhere-Hyohyobako," a non-infrastructure-dependent sensor network based on BLE advertising. Future work includes field testing to evaluate observation accuracy and feasibility of social implementation, with the goal of establishing a new weather observation infrastructure that connects academia, public sectors, and citizen communities.

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